

236502 – Artificial Intelligence Project

Final Presentation

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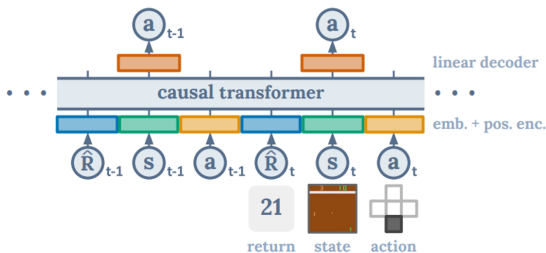
Problem & Approach

Problem & Approach Recap

- **Research Question:** *Can DT overcome RL in energy environment?*
- **Approach Overview:**
 - *Build and train RL and DT agents in ANM6easy in order to compare between the different architectures.*
 - *Data sources: Decision Transformer: Reinforcement Learning via Sequence Modeling, ANM6easy paper in gym-anm.com*

DT architecture

- **Transformer:** Neural-Network combine MLP with attention layers in order to learn the connections between the data's tokens.
- **Decision Transformer:**
 - A transformer adaptation for RL problems.
 - When receiving data sequences, split it into "words" made of {state, action, rtg} and decide which action to use next.
 - During the train process the DT learns the environment and when given a sequence of a run and a target reward can decide which is the most suitable action.



- **Network contains:**
 - 6 buses
 - 3 passive loads: industrial, residential, and EV charging
 - 2 renewable energy sources: wind and PV plant
 - 1 storage unit
 - 1 fossil fuel generator
- **State space:**
 - Active and reactive power of all 7 devices
 - Max active power of the 2 generators
 - Storage level
 - Time index
 - Each state is an 18-dimensional vector
- **Action space:**
 - Change in active power of 6 devices (all except buses)
 - Each action is a 6-dimensional vector
- **Reward space:**
 - Penalty + energy loss
 - Each reward is a scalar in $[-200, 0]$

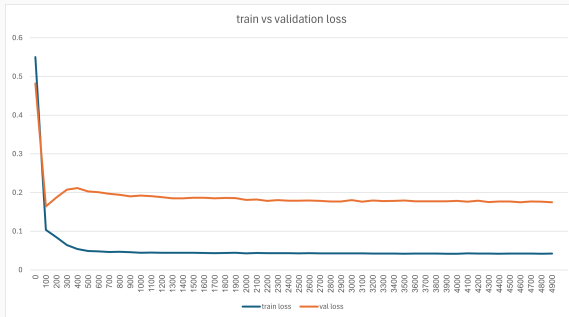
Training process

Hyper parameters & training

- We trained several decision transformers (DTs) with different hyperparameters and evaluated their performance to determine the most successful one.
- The selected DT has 6 layers, 8 heads, a learning rate of 0.0001, and was trained for 5000 epochs.
- Although it did not achieve the lowest validation loss (0.174 vs. 0.164), it is still the best-performing DT we trained.
- This can be explained by the fact that, although the validation loss does not decrease monotonically, it consistently hovers around 0.17.
- In general, the DT architecture was trained on a large dataset, which helps prevent overfitting and allows it to generalize effectively.

Loss function

- train and validation loss:



As expected, the training loss decreases monotonically. However, the validation loss stabilizes around 0.17. During the evaluation phase, we tested several different decision transformers (DT) - both against each other and against PPO - and the most successful DT trained for 5000 epochs.

Key Results

Key Results

- **Evaluation Method:** *running all the agents in the environment, Each time they all start from the same random state.*
- **Notable Outcomes / Demos:**
 - *the DT can be as good as the RL ppo agent*
 - *Although we expected it to be better then the ppo agent, they have very similar reward mean in 1000 runs.*
 - *Based on an evaluation test from 05.07.*

Evaluation

Evaluation

- **Success Criteria Met?** *At this moment, although the DT do succeeded in making good decisions, it is not better then the RL.*
- **Quantitative Reflection:**
max episode length: 480, episodes: 1000
DT stats:
mean reward for step: -0.077
max cumulative reward: -22.378
min cumulative reward: -54.451
PPO stats:
mean reward for step: -0.068
max cumulative reward: -19.489
min cumulative reward: -47.214
- **Qualitative Reflection:** *the longer the run (episode) the better the DT*

Takeaways & Future Work

Takeaways & Future Work

- **Key Lessons Learned:**
 - *the DT might have an advantage on longer runs*
 - *In order to use the DT's "non policy" upside, we need to use an environment with a complex reward-space*
- **Potential Improvements:**
 - *Building DT that can run long episodes and test it*
- **If We Had More Time:**
 - *Building DT that can run long episodes*
 - *Using more RL agents*
 - *Research in a different environment*